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# Statistical Models
# Week 5 : general results and methods for glm

comments

How do we estimate parameters in a generalized linear models? How do we test hypotheses about the parameters?

We will continue with general results and techniques for glms, in particular maximum likelihood estimation, but also look at fitting glms practically using R. Goodness of fit of glm can be assessed by the deviance – a likelihood ratio statistic ( ) – or by studying residuals. We will also see a brief overview of the iterative reweighted least squares technique to find maximum likelihood estimators, and have a look at the asymptotic properties of MLE in the context of glm. If time allows, we will turn towards special models, in particular proportion data.

## Relevant Readings

Davison, 10.2, 10.3.2, evtl. 10.4; supplemented by Agresti, 4.4.6, 4.5.

## Lecture Material

- Link to onenotes and cheat sheet on Google drive
- "My first glm" Try fitting a binomial proportion glm on the butterfly data
- Analysis of a glm Residuals and model reduction in R

## Exercises 5

**Problem 5.1:** GLM for comparing two groups

(Compare Agresti 2015, Exercises 4.6 and 4.7)
In this exercise we consider data from two groups $A$ and $B$ of size $n_A$ and $n_B$, that can be modelled by i.i.d. Poisson variables within each group (thus we have in all a sample of size $n = n_A + n_B$). The first $n_A$ observations belong to a group $A$, and the remaining $n_B$ belong to a group $B$, with different group means $\mu_A$ and $\mu_B$. For the analysis by a GLM, we use the linear predictor $g(\mu_i) = \eta_i = \beta_0 + \beta_1 \mathbf{1}(i \in \{1, ..., n_A\})$.

a. Show that for the log-link function, the GLM likelihood equations imply that the fitted means $\hat{\mu}_A$ and $\hat{\mu}_B$ equal the sample means.

b. Show that the result of a) holds no matter what link function $g$ is used or what distribution for the $Y_i$ is assumed.

You could also try to generalize the result to more than two groups.

**Problem 5.2:** Survival times

Times to event (survival times) are often modeled by an exponential distribution. Recall that this is a special case of Gamma distribution :-)

The following data example deals with survival data of leukemia patients and is a classic [a]. Here $y_i$ represent times to death in weeks from diagnosis, and $x_i$ the $\log_{10}$ of initial white blood cell count, for seventeen patients.
The R code below generates vectors `y` and `x`:

```
y <- c(65, 156, 100, 134, 16, 108, 121, 4, 39, 143, 56, 26, 22, 1, 1, 5, 65)
x <- c(3.36, 2.88, 3.63, 3.41, 3.78, 4.02, 4.00, 4.23, 3.73, 3.85, 3.97, 4.51,
       4.54, 5.00, 5.00, 4.72, 5.00)
```

a. Plot $y_i$ against $x_i$. Do the data show any trend?

b. A possible specification for $\mathrm{E}\, y_i$ is

$$\mathrm{E}\,(y_i) = \exp(\beta_1 + \beta_2 x_i),$$

which will ensure that $\mathrm{E}\,(y)$ is non-negative for all values of the parameters and all values of $x$. Which link function is appropriate in this case?

c. Using R's function `glm`, fit a model that assumes that the $y_i$ have an exponential distribution, with mean $\mathrm{E}\,(y_i)$ given in (b). Hint: use the Gamma family. The Gamma shape parameter has no influence on the estimate of the model parameters in a glm.

d. For the fitted model, compare the observed values $y_i$ and fitted values $\hat{y}_i = \exp(\hat{\beta}_1 + \hat{\beta}_2 x_i)$, and use the standardized residuals $r_i = (y_i - \hat{y}_i)/\hat{y}_i$ to investigate the adequacy of the model.
(Note: $\hat{y}_i$ is used as the denominator of $r_i$ because it is an estimate of the standard deviation of $y_i$— why is this?)

[a] data from the book Cox & Snell (1981) Applied statistics, found in Dobson & Barnett (2008): An Introduction to Generalized Linear Models

**Problem 5.3:** O-ring failure in space shuttles: glm and model checking

Davison, Practicals 10.1.
The `shuttle` data are unfortunately slightly corrupted – they have row names that are numbers, but they should be character.
Repair them by running

```
row.names(shuttle) <- as.character(row.names(shuttle))
```

or simply remove them with `row.names(shuttle) <- NULL` before doing any analysis.

**Problem 5.4:** Deviance goodness of fit test for the Poisson model - when does asymptotic theory work, and when not?

In the lectures, we have learned that the scaled deviance can be used to test if a model describes the data well, since scaled deviance between a model and the corresponding saturated model often has an asymptotic $\chi^2$-distribution. In this exercise, you explore this for a Poisson model by simulation. Let $Y_1, \ldots, Y_n$ be independent Poisson distributed variables with $\mathrm{E}\, Y_i = \lambda_i$. This is the saturated model $M_0$. In the submodel $M_1$, all mean values coincide, that is, $\lambda_1 = \cdots = \lambda_n$.

a. Show that the scaled deviance for model $M_1$ is given as

$$D = 2\sum_{j=1}^{n} Y_j \log\left(\frac{Y_j}{\bar{Y}}\right), \quad \bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i,$$

where we set $Y_j \log(Y_j/\bar{Y}) = 0$ if $Y_j = 0$.

Extra (ad libitum): show that $\lim_{x\to 0} x\log(x) = 0$.

b. Simulate $N = 1000$ values of $D$ and compare with a $\chi^2$-distribution with $n-1$ degrees of freedom. For a visual comparison, you can for example use an appropriate qq-plot (not qqnorm). Also estimate the probability that $D$ exceeds the $95\%$-quantile of the $\chi^2_{n-1}$-distribution.

Do this for several values of $\lambda$ and $n$, e.g. $\lambda = 1, 10, 100$ and $n = 10, 100, 1000$. In some cases, you will observe that the $\chi^2$-approximation is not good. Summarize your findings - when would you recommend a goodness-of-fit test based on the $\chi^2$-approximation of the scaled deviance, and when not?

Try to explain your findings.